U-Hub

Turning Used Cooking Oil into Traceable Circular Feedstock

Solving urban feedstock recovery inefficiency through traceable supply systems

Our Team

U-Hub

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System Architecture, Operations & Data

Business modeling meets operational design — built for real-world constraints.

Based in Surabaya, Indonesia

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Situation Overview

HIGH VOLUME, BUT ZERO SYSTEM INTEGRATION

52+
clusters

1,100+
vendors

- 1 L/vendor/day
- → **200,000 L/year (Year 1 addressable, 3-5 pilot clusters)**
- ~$0.50/L market value
- → **$100K/year recoverable value (Year 1)**

(Assumption: ~60% frying vendors, 1 L/day conservative, 300 operating days. Year 1 = 3–5 pilot clusters only.)

SYSTEM ISSUES ⚠️

fragmented supply | no traceability | no incentive

S: Scalable density, asset-light, existing demand (SAF, biodiesel)	**W:** Supply inconsistency (highly real), dependency on 3rd-party-logistics
O: ESG mandates	**T:** Informal collector competition on payout price

S
W
O
T

Problem Statement

No system → No accountability → No market

OPERATIONAL

No structured micro-collection

Only 26% of F&B businesses in Indonesian cities have any waste management program (Gunawan, 2021)

ENVIRONMENTAL

Oil systematically dumped into drainage

UCO raises wastewater BOD/COD levels 50–100x above safe limits, causing aquatic ecosystem damage

ECONOMIC

~$0.50/L value permanently lost

Surabaya's 52+ clusters alone represent ~$165K/year in unrecovered feedstock value

This is not a supply shortage. This is a system coordination failure.

Assessment of Alternatives

Evaluated on upfront cost, traceability, fit for dense small-vendor clusters, and potentials of Green Circularity.

Approach	Upfront CapEx	Traceability & QC	Fit for Small Stalls	Green Circularity Potential	Selected?
1. IoT Smart Bins	High ✗ (Costly scale)	High ✓	Low ✗	High ✓	**No**
2. Traditional Collectors	Low ✓	Low ✗ (Contamination risk)	Med – (Weak control)	Low ✗	**No**
3. U-Hub (Asset-Light)	**Low ✓**	**High ✓**	**High ✓**	**High ✓**	**YES**

Requires vendor behavior change and buyer relationship to be established

No existing approach solves cost efficiency and traceability simultaneously at small-vendor scale.

Solution → U-Hub: Digital Chain-of-Custody System

Vendor → **U-Hub**
Vendor App, Driver App, Internal Dashboard
→ **Industry**

OPERATIONAL DETAILS

- Volume recorded
- Vendor ID logged
- Oil condition checked
- Exportable batch records for Industrial Buyer Compliance

WHY IT WORKS

- Converts fragmented supply into structured, documented input
- Enables compliance-ready traceability for industrial buyers (ISCC standard)
- Low asset intensity — no depot, no owned vehicles required

Implementation

"Designed for density, not distance"

ROUTING

- 1 cluster = 20–40 vendors
- Scheduled routing
- Cost efficiency via density (~30–60 L per scheduled pickup per cluster visit)

QUALITY CONTROL

- On-site physical check (every pickup)
- Batch lab test triggered by contamination flag only

CONTINGENCY PLAN

RISK	RESPONSE
Low Vendor Participation?	Onboard via site manager incentive; pivot to hotel & franchise chains for higher volume per point
UCO buy price increases due to collector competition?	Margin maintained through route efficiency — cost per liter drops as volume scales
Unconfirmed industrial buyer at Month 9?	Bridge via mid-tier aggregator while pursuing direct offtake from biodiesel processor

UNIT ECONOMICS

- Sell: $0.50
- Buy: $0.30
- Logistics: $0.13
- **Net: $0.07/L**

Margin maintained through route efficiency and cluster density — not price premium

FINANCIALS

- CapEx: $18,200
- Monthly cost: $2,200
- Break-even: Month 9

IMPACT

- 200,000 L/year
- 1L ≈ 2.5 kg CO_2e

500 tons **CO_2e/year**

(Argonne GREET lifecycle model)

Metric	Year 1 (Pilot)	Year 2 (Expansion)	Year 3 (Network)
Active Clusters	3 - 5 Clusters	12 - 15 Clusters	**25+ Clusters**
Annual Volume	~200,000 L	~850,000 L	**~2.1 Million L**
Annual Revenue	~$100,000	~$425,000	**~$1.05 Million**
EBITDA Profile	**Pilot Phase**	**Improving Unit Economics**	**Path to Scaled Profitability**

The Moat

Auditable chain-of-custody data becomes a defensible advantage as our network scales.

Impact (SDG 11, 12)

Equivalent to removing ~108 passenger cars from the road for one year. *(*)*

The Vision

Building the trust, traceability, and recovery layer for the green urban food economy.

(*) Based on lifecycle emission comparison: UCO biodiesel ~20 g CO_2e/MJ vs fossil diesel ~95 g CO_2e/MJ. Source: Argonne National Laboratory GREET model

Conclusion

Solves the collection coordination failure

500 tons CO_2e diverted annually (Year 1)

~$100K recoverable value per year (Year 1)

"U-Hub is the infrastructure layer for circular urban economies"

Thank you · Terima kasih · ขอบคุณ

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